

CIFAD: Causal-Invariant Subspace Learning for Few-Shot Anomaly Detection on Dynamic Relational Graphs

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Abstract

Abnormal user detection has been a critical and widely studied research problem in social networks since these users can create significant risks to platform security and privacy leakage. Currently, graph-based models are commonly used for exploring the structured social network data and temporally dynamic user interactions, leading to significant advances in dynamic heterogeneous graph-based abnormal user detection. However, most existing approaches are correlation-driven and lack the ability to separate stable patterns from transient noise. Furthermore, these methods are highly dependent on inherent labels and fail to detect common few-shot anomalies in social networks. To address these issues, we propose CIFAD, a Causal-Invariant Few-shot Anomaly Detection method that improves few-shot anomaly detection with an active annotation strategy. Specifically, CIFAD first integrates a sparse lagged attention encoder to model multi-relational temporal interactions. Furthermore, it introduces causal-invariant subspace decomposition to disentangle stable causal signals from dynamic environmental noise and improve generalization. Finally, it designs an active annotation strategy based on influence functions and coverage optimization to maximize the utility of limited labels in a closed-loop process. Extensive experiments on multiple real-world social network datasets demonstrate that our method achieves higher accuracy than state-of-the-art methods, validating its robustness in abnormal user detection for social networks.

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CCS Concepts

• **Computing methodologies** → *Learning paradigms*; **Causal reasoning and diagnostics**; • **Security and privacy** → **Intrusion/anomaly detection and malware mitigation**; • **Mathematics of computing** → **Graph algorithms**; • **Theory of computation** → **Social networks**.

Keywords

Graph-based Anomaly Detection, Social Networks, Dynamic Graphs, Few-Shot Learning, Causal-Invariant Representation

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1 Introduction

As online social networks grow in scale and societal influence, safeguarding their security, reliability, and trustworthiness has become increasingly critical. Among the associated challenges, abnormal-user detection plays a fundamental role in maintaining platform integrity [23]. Malicious accounts can disseminate misinformation, leak private data, and conduct fraud, thereby undermining user trust and platform integrity [24]. In real deployments, missed detections may trigger misinformation cascades and substantial financial losses, whereas excessive false alarms increase human review burden and harm user experience. This problem is further complicated by continuously evolving network dynamics, heterogeneous interaction patterns, and limited annotation budgets, which together make robust and efficient anomaly detection a long-standing challenge [27]. Consequently, practical solutions must remain reliable under distribution shifts and scarce supervision [26].

Graph-based models have emerged as a powerful approach for this task, as they naturally capture the relational structure of social interactions [25, 39]. Research in this area has evolved from early methods that analyzed static graph snapshots to dynamic graph neural networks that model temporal evolution of interactions [17]. More recently, dynamic heterogeneous graph models have further improved performance by incorporating different types of relations such as follows, replies, or likes, together with their temporal patterns, offering a more comprehensive view of user behavior [26, 30].

Despite this progress, current methods still face two major limitations that affect their real-world applicability [37]. Firstly, most models are correlation-driven. They learn patterns based on statistical associations but fail to separate the true causes of anomalies from coincidental, environment-specific noise [2]. For example, a model might associate high posting frequency with malicious activity simply because both often occur during popular events [42]. However, this correlation is misleading since normal users also tend to post frequently during holidays or breaking news periods [4]. When social network conditions change, models relying on such transient correlations tend to perform poorly, as they lack an understanding of stable abnormal behavior [5]. Secondly, anomaly detection suffers from a severe shortage of labeled data [9]. Identifying abnormal accounts requires costly expert review, resulting in only a small number of confirmed samples for training [37]. This few-shot setting makes it difficult for complex models to learn effectively, often causing them to overfit the limited labeled data and perform poorly on unseen anomalies [7]. These two issues, namely model instability and label scarcity, interact in a mutually reinforcing manner, which further increases the difficulty of achieving robust detection [35].

To address these challenges, we propose CIFAD, a causal-invariant few-shot anomaly detection method for dynamic social networks. CIFAD learns robust representations by combining temporal heterogeneous graph modeling, causal-invariant signal isolation, and label-efficient active annotation. Extensive experiments on real-world datasets demonstrate consistent improvements in accuracy and robustness over state-of-the-art methods. Our main contributions are as follows:

- We propose the CIFAD method, a causal-invariant few-shot anomaly detection method that overcomes the limitations of correlation-driven models and label dependency.
- We design a sparse lagged attention encoder to effectively model multi-relational temporal interactions in dynamic social networks.
- We introduce a causal-invariant subspace decomposition mechanism to disentangle stable causal signals from dynamic environmental noise, improving generalization and robustness.
- We develop an active annotation strategy based on influence functions and coverage optimization to maximize the utility of limited labels in a closed-loop learning process.

2 Related Work

2.1 Static Graph Anomaly Detection

In static graphs, anomaly detection aims to identify nodes, edges, or substructures that significantly deviate from the dominant graph

pattern without considering temporal evolution [8, 21]. Early approaches primarily relied on structural reconstruction or attribute reconstruction to capture abnormal patterns, but recent advances have shifted toward explainable, spectral, and contrastive paradigms to improve both detection accuracy and interpretability [14].

Yang et al. [32] proposed GRAM, an interpretable framework that leverages gradient-based attention maps to highlight anomalous subregions in graphs, thereby enhancing transparency and trustworthiness in node-level detection. Their results show that interpretability mechanisms can contribute to more stable and reliable anomaly reasoning. Building on interpretability, Zheng et al. [40] introduced Dynamic Spectral Graph Anomaly Detection, which analyzes variations in spectral energy to capture abnormal structural signals even in static snapshots. To further exploit graph topology, Zheng et al. [41] developed CARE, a cluster-aware multi-view anomaly detector that integrates pseudo-label information into the adjacency matrix to capture both local and global node affinities. By introducing a similarity-guided regularization, CARE mitigates pseudo-label bias and improves generalization in multi-view scenarios. Li et al. [10] proposed UMGAD, an unsupervised multiplex graph anomaly detection model that couples graph-masked autoencoders with contrastive learning to capture both attribute- and structure-level inconsistencies across heterogeneous relations. It further designs an adaptive thresholding mechanism for robust scoring in fully unsupervised environments. More recently, Niu et al. [16] presented Zero-shot Generalist Graph Anomaly Detection (Z-GGAD), which employs unified neighborhood prompts to achieve cross-domain generalization without retraining. By aligning representations through prompt-based adaptation, Z-GGAD demonstrates strong zero-shot transferability across diverse static graph datasets, highlighting the potential of prompt learning in anomaly detection.

Despite their effectiveness in identifying structural irregularities, the above methods remain inherently correlation-driven, as they rely on feature reconstruction or similarity-based reasoning [1].

2.2 Dynamic Graph Anomaly Detection

In dynamic social networks, nodes and edges evolve over time, making anomaly detection more challenging due to shifting structures and temporal dependencies [20, 38]. Dynamic graph anomaly detection (DGAD) therefore focuses on identifying irregularities that emerge or persist across temporal snapshots [22, 33]. Unlike static settings, DGAD jointly model spatial relationships and temporal evolution to ensure stable and adaptive representation learning [34].

Tian et al. [19] proposed SAD, a semi-supervised method that combines a time-aware memory bank with pseudo-label contrastive learning to leverage unlabeled temporal data. However, it remains sensitive to short-term correlations. Yuan et al. [36] introduced MADG, a motif-level detector that employs motif-augmented graph convolution and temporal self-attention to capture the evolution of motif structures over time. Guo et al. [6] developed RustGraph, which jointly learns structural-temporal representations through a variational graph auto-encoder with fine-grained contrastive learning, improving robustness against noisy labels. Yang et al. [31] proposed GeneralDyG, which samples temporal ego-graphs to extract spatial and temporal features, enhancing adaptability across

evolving environments. Recently, Liu et al. [13] presented AnomalyLLM, integrating large language models with temporal graph encoders for few-shot edge-level anomaly detection, demonstrating the potential of semantic priors in dynamic modeling.

Overall, most studies remain correlation-driven and lack causal interpretability. They struggle to disentangle stable causal patterns from transient noise, limiting robustness under distributional shifts.

2.3 Few-shot Anomaly Detection

Few-shot anomaly detection (FSAD) aims to identify abnormal samples when only a few labeled anomalies are available. This task is challenging because anomalies are rare and diverse, while effective detection typically requires sufficient supervision [29]. To mitigate data scarcity, recent studies have developed label-efficient paradigms that transfer meta-knowledge, adapt pre-trained models, or generate synthetic anomalies for better generalization [16]. Overall, these approaches can be grouped into three categories: (1) reconstruction-based methods that rebuild query features using few normal samples, (2) adaptation-based methods that fine-tune pre-trained representations for domain alignment, and (3) prompt- or meta-learning methods that leverage transferable priors to enhance anomaly discrimination [15, 28].

Fang et al. [3] proposed FastRecon, which reconstructs query samples from few-shot normal references and identifies anomalies by measuring reconstruction discrepancies. Liao et al. [12] introduced Coft-AD, a contrastive fine-tuning method that aligns pre-trained source models through cross-instance contrastive learning. Li et al. [11] presented PromptAD, which synthesizes pseudo-anomaly prompts via semantic concatenation to refine decision boundaries using only normal data. Tao et al. [18] further proposed KAG-Prompt, a kernel-aware prompt learning framework that models hierarchical contextual relations for few-shot detection. These methods substantially improve anomaly detection with minimal supervision but remain largely correlation-driven, focusing on reconstruction or representational similarity while neglecting causal invariance. Consequently, their generalization often degrades under dynamic or heterogeneous settings.

As the above review indicates, a significant gap remains for a unified approach that is both causally robust to environmental changes and effective under limited supervision. In this work, we aim to bridge this gap by designing a robust few-shot anomaly detection method in dynamic social networks.

3 Problem Formulation

We model a social network as a sequence of dynamic heterogeneous graphs $G = \{G_t\}_{t=1}^T$, where $G_t = (V, E_t, X_t, R)$ denotes the snapshot at time t . Given a small labeled set $L = \{(v, t, y_{(v,t)})\}$ with $y_{(v,t)} \in \{0, 1\}$ and $|L| \ll |V| \cdot T$, our goal is to learn a predictor that estimates the anomaly probability of node v at time t from its historical context: $\hat{y}_{(v,t)} = f(v, t \mid G_{\leq t})$, where $G_{\leq t} = \{G_1, \dots, G_t\}$. For causal-invariance regularization, we further partition the temporal domain into pseudo-environments $\mathcal{E}$ via fixed-window segmentation or topic-aware clustering, which does not rely on anomaly labels.

4 Method

4.1 Overview

Figure 1 presents an overview of CIFAD, a closed-loop method for few-shot anomaly detection in dynamic heterogeneous social networks. CIFAD consists of three tightly coupled modules: (1) a lag-aware heterogeneous temporal encoder for modeling multi-relational dependencies, (2) a causal-invariant subspace decomposition that separates stable behavioral signals from environment-specific perturbations, and (3) an active annotation strategy that selects informative samples to maximize supervision efficiency. These components interact iteratively, progressively refining invariant representations and improving robustness under distribution shifts with scarce labels.

4.2 HGTT-SLA Encoder

The first component of CIFAD is a dynamic heterogeneous graph encoder designed to model asynchronously evolving, multi-relational interactions. To capture both multi-scale structure and long-range temporal dependencies, we introduce a Heterogeneous Graph Temporal Transformer with Sparse Lagged Attention (HGTT-SLA). HGTT-SLA extends temporal GNNs by explicitly incorporating relation-specific temporal lags and imposing sparse attention, which improves robustness and reduces overfitting.

Formally, for each node v at time t , HGTT-SLA aggregates messages from temporal neighbors spanning multiple relation types and time lags:

$$\mathbf{h}_{(v,t)} = \text{HGTT-SLA}(v, t; \{(u, r, \tau) \mid (u, t - \tau) \rightarrow (v, t) \in \mathcal{E}_t\}). \quad (1)$$

Here, each tuple (u, r, τ) specifies a temporal interaction from node u to node v with relation type r and lag τ . This formulation enables HGTT-SLA to model heterogeneous temporal dependencies over long horizons.

For each interaction, we construct relation-aware query, key, and value projections:

$$\begin{aligned} \mathbf{q}_{(v,t)} &= W_Q x_{(v,t)}, \\ \mathbf{k}_{(u,t-\tau)}^{(r)} &= W_K^{(r)} x_{(u,t-\tau)}, \\ \mathbf{v}_{(u,t-\tau)}^{(r)} &= W_V^{(r)} x_{(u,t-\tau)}, \end{aligned} \quad (2)$$

where $x_{(v,t)}$ denotes the node feature, W_Q is shared across relation types, and $W_K^{(r)}, W_V^{(r)}$ are relation-specific transformations. The attention coefficient from node $(u, t - \tau)$ to (v, t) is computed as:

$$\alpha_{u \rightarrow v}^{(r,\tau)}(t) = \text{softmax}_{u \in \mathcal{N}_{r,\tau}(v,t)} \left(\frac{\mathbf{q}_{(v,t)}^\top \mathbf{k}_{(u,t-\tau)}^{(r)}}{\sqrt{d}} \right) \quad (3)$$

where $\mathcal{N}_{r,\tau}(v, t)$ denotes the set of temporal neighbors connected to (v, t) under relation r with lag τ , and d is the hidden dimension used for scaling. We then aggregate messages within each relation-lag group as:

$$\mathbf{m}_{(v,t)}^{(r,\tau)} = \sum_{u \in \mathcal{N}_{r,\tau}(v,t)} \alpha_{u \rightarrow v}^{(r,\tau)}(t) \mathbf{v}_{(u,t-\tau)}^{(r)}. \quad (4)$$

To avoid attending indiscriminately to all relation-lag combinations, we introduce learnable non-negative gating weights $g_{r,\tau}$ that

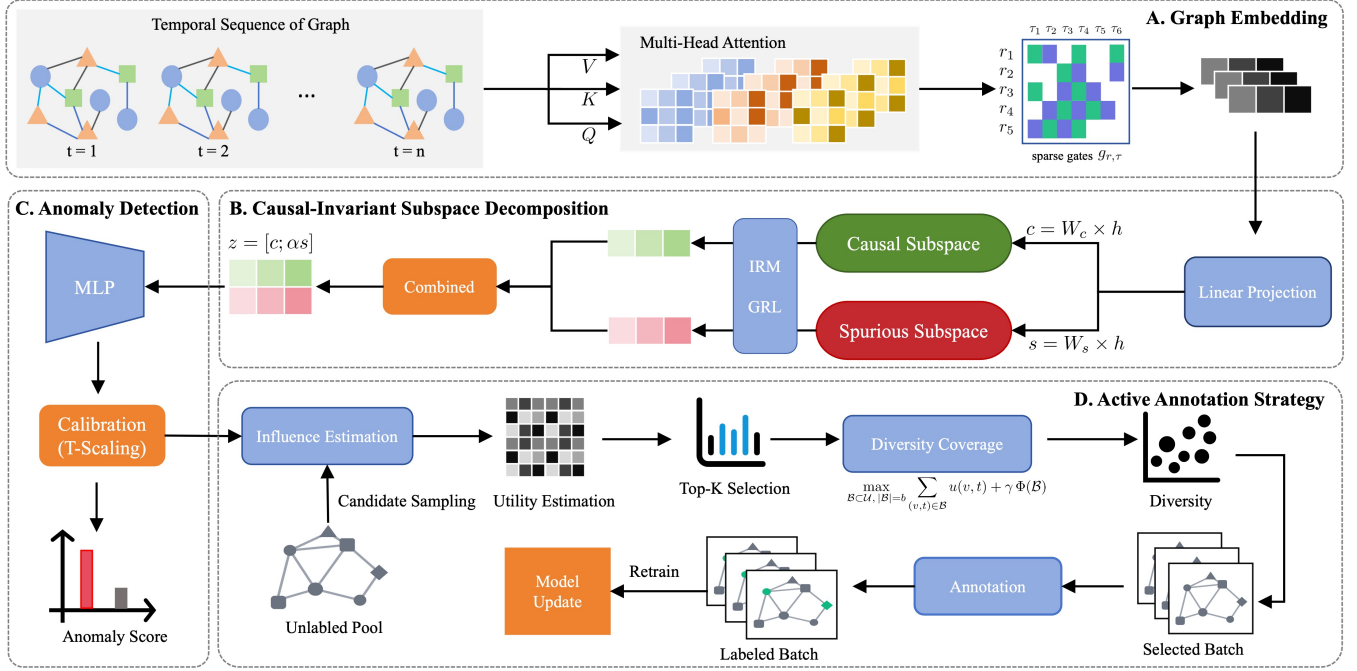


Figure 1: Overall architecture of CIFAD. The method integrates (1) Sparse Lagged Attention Encoding, (2) Causal-Invariant Subspace Decomposition, and (3) Active Annotation, forming a closed-loop causal learning process for few-shot anomaly detection in dynamic social networks.

adaptively control the contribution of each pair (r, τ) :

$$\tilde{\mathbf{h}}_{(v,t)} = \sum_{r \in \mathcal{R}} \sum_{\tau \in \mathcal{T}} g_{r,\tau} \mathbf{m}_{(v,t)}^{(r,\tau)}, \quad \mathbf{h}_{(v,t)} = W_O \tilde{\mathbf{h}}_{(v,t)}, \quad (5)$$

where W_O is an output projection. This gating mechanism emphasizes informative relation-lag pairs while suppressing redundant or noisy dependencies.

To regularize the gating parameters, we apply a structured sparsity term that favors compact and interpretable structure:

$$\mathcal{L}_{\text{spa}} = \lambda_r \sum_{r \in \mathcal{R}} \|g_{r,:}\|_2 + \lambda_\tau \sum_{\tau \in \mathcal{T}} \|g_{:, \tau}\|_1, \quad (6)$$

where λ_r and λ_τ control the levels of group and element sparsity, respectively. This mixed-norm constraint enables HGTT-SLA to adaptively select the most causally informative relation-lag channels, thereby optimizing interpretability and efficiency.

By integrating heterogeneous attention, temporal lag modeling, and sparsity gating, HGTT-SLA produces contextualized embeddings $\mathbf{h}_{(v,t)} \in \mathbb{R}^d$ that effectively capture long-range, multi-relational temporal dynamics.

4.3 Causal-Invariant Subspace Decomposition

Although the HGTT-SLA encoder generates expressive temporal embeddings, these representations inevitably mix two types of information: stable causal factors that consistently govern user behaviors and transient environmental noise arising from event bursts, topic drift, or platform fluctuations. To achieve robust anomaly detection under few-shot supervision, CIFAD introduces a *causal-invariant*

subspace decomposition that explicitly disentangles these components, isolating invariant mechanisms from environment-specific variations.

Formally, for each node v at time t , the encoder output $\mathbf{h}_{(v,t)}$ is projected into two latent subspaces:

$$\mathbf{c}_{(v,t)} = W_c \mathbf{h}_{(v,t)}, \quad \mathbf{s}_{(v,t)} = W_s \mathbf{h}_{(v,t)}, \quad (7)$$

where $\mathbf{c}_{(v,t)}$ denotes the causal subspace capturing stable, environment-agnostic signals, and $\mathbf{s}_{(v,t)}$ models short-term, context-dependent fluctuations. The objective is to ensure that $\mathbf{c}_{(v,t)}$ encodes only causally consistent information, while $\mathbf{s}_{(v,t)}$ absorbs environment-specific variability.

To achieve this disentanglement, CIFAD adopts three complementary regularization objectives. **(1) Adversarial deconfounding.** We introduce an environment discriminator D_e to encourage the causal representation $\mathbf{c}_{(v,t)}$ to be environment-invariant. Specifically, D_e is trained to predict the pseudo-environment label e from $\mathbf{c}_{(v,t)}$ by minimizing

$$\mathcal{L}_{\text{adv}} = -\mathbb{E}_{(v,t)} [\log D_e(\mathbf{c}_{(v,t)})]. \quad (8)$$

Meanwhile, the encoder is updated through a gradient reversal layer, which effectively maximizes $\mathcal{L}_{\text{adv}}$ with respect to the encoder parameters and thus removes environment-discriminative information from $\mathbf{c}_{(v,t)}$.

(2) Invariant risk minimization. To enforce predictive consistency across pseudo-environments, we adopt the principle of Invariant Risk Minimization (IRM). For each environment e , the classification loss $\mathcal{L}_{\text{cls}}^{(e)}$ is computed on samples within $\mathcal{D}^{(e)}$, and

the optimal classifier is constrained to yield identical gradients across environments:

$$\mathcal{L}_{\text{IRM}} = \sum_e \left\| \nabla_w \mathbb{E}_{(v,t) \in \mathcal{D}^{(e)}} [\ell(w^\top c_{(v,t)}, y_{(v,t)})] \right\|_2^2, \quad (9)$$

where w denotes the classifier weights. This constraint stabilizes the decision boundary of the classifier trained on $c_{(v,t)}$ and prevents it from overfitting to any single environment.

(3) Orthogonality constraint. To promote a disentangled factorization between the causal and spurious components, we impose an orthogonality penalty:

$$\mathcal{L}_\perp = \|W_c^\top W_s\|_F^2, \quad (10)$$

where W_c and W_s denote the projection matrices for the causal and spurious subspaces, respectively. This regularizer penalizes cross-subspace correlation to minimize information leakage and encourage non-overlapping representations.

Together, these regularization objectives yield a causal subspace $c_{(v,t)}$ that captures stable and transferable anomaly-related patterns, while the spurious subspace $s_{(v,t)}$ absorbs transient and non-stationary variations. This decomposition is particularly important under few-shot settings, where the decoder must generalize from limited labels while remaining reliable under environment-dependent perturbations. The disentangled causal representation $c_{(v,t)}$ is subsequently used for anomaly detection and probability calibration, forming the basis for reliable inference under temporal distribution shifts.

4.4 Anomaly Detection and Probability Calibration

After disentangling the causal and spurious subspaces, CIFAD trains a classifier to detect whether node v exhibits anomalous behavior at time t . The causal subspace $c_{(v,t)}$ provides stable, environment-invariant evidence, whereas the spurious subspace $s_{(v,t)}$ can retain transient contextual signals that offer complementary information for detection. Therefore, the two representations are combined through a weighted concatenation:

$$z_{(v,t)} = [c_{(v,t)}; \beta s_{(v,t)}], \quad (11)$$

where β is a learnable scaling factor controlling the influence of $s_{(v,t)}$. By tuning β , the detector can rely primarily on invariant causal cues and incorporate short-term context only when beneficial.

The anomaly probability is predicted by a nonlinear classifier $f_\theta(\cdot)$ parameterized by θ :

$$\hat{y}_{(v,t)} = \sigma(f_\theta(z_{(v,t)})), \quad (12)$$

where $\sigma(\cdot)$ denotes the sigmoid function. To address the inherent class imbalance between normal and anomalous samples, CIFAD employs either a weighted binary cross-entropy loss or a focal loss, depending on dataset characteristics:

$$\mathcal{L}_{\text{cls}} = \text{BCE}(Y, \hat{Y}) \quad \text{or} \quad \mathcal{L}_{\text{cls}} = \text{Focal}(Y, \hat{Y}). \quad (13)$$

These objectives amplify gradients for rare anomalies, improving recall and training stability under limited supervision.

In practice, neural detectors often exhibit overconfident outputs, reducing their reliability in active learning and downstream

analysis. To mitigate this, CIFAD introduces a temperature-based calibration mechanism:

$$\hat{p}_{(v,t)} = \sigma\left(\frac{f_\theta(z_{(v,t)})}{T_{\text{cal}}}\right), \quad (14)$$

where T_{cal} is a learnable temperature parameter optimized on a held-out validation set. By adjusting the sharpness of predicted logits, this step aligns model confidence with the true likelihood of anomalies, producing well-calibrated probability estimates interpretable as risk scores.

Ultimately, this module outputs calibrated anomaly probabilities $\hat{p}_{(v,t)}$ for all node–time pairs, offering a reliable and interpretable measure of abnormality that supports both robust detection and the subsequent active annotation process.

4.5 Active Annotation Strategy

In few-shot anomaly detection, the scarcity of labeled anomalies remains a core challenge, as supervised models often underperform when annotation budgets are limited. Although CIFAD can learn causal-invariant representations, its performance still depends on access to a small yet strategically selected set of labeled samples. To better utilize limited supervision, we design an active annotation strategy that continuously selects the informative instances for labeling and retraining in a closed-loop manner. This mechanism enables CIFAD to allocate annotation efforts efficiently and achieve steady improvements under constrained labeling resources.

The key idea is to quantify how much each unlabeled instance is expected to contribute to model improvement. Given a trained model parameterized by θ , we estimate the impact of labeling an unlabeled node–time pair (v, t) using influence functions. The resulting influence score reflects the first-order effect of incorporating that sample into the training process:

$$\mathcal{I}(v, t) \approx g^\top H^{-1} \nabla_\theta \ell_{(v,t)}, \quad (15)$$

where H is the Hessian matrix of the training loss with respect to θ , and $g = \nabla_\theta \mathbb{E}_{\text{val}}[\ell]$ denotes the gradient on a small validation set. Intuitively, $\mathcal{I}(v, t)$ reflects how much the model parameters would change if (v, t) were annotated and used for retraining—samples with larger absolute influence are expected to yield greater performance gains once labeled.

Selecting samples solely based on their influence may cause redundancy, as many high-influence candidates can correspond to similar behaviors or temporal contexts. To ensure diversity among queried samples, we incorporate a coverage-aware selection criterion. Specifically, given the unlabeled node pool $\mathcal{U}$, we seek to select a batch $\mathcal{B} \subset \mathcal{U}$ of size b that jointly maximizes informativeness and diversity:

$$\max_{\mathcal{B} \subset \mathcal{U}, |\mathcal{B}|=b} \sum_{(v,t) \in \mathcal{B}} \mathcal{I}(v, t) + \gamma \Phi(\mathcal{B}), \quad (16)$$

where $\Phi(\mathcal{B})$ is a coverage function that encourages diversity among selected samples, and γ balances the trade-off between informativeness and coverage.

This formulation ensures that the annotation budget is effectively allocated to samples that are both influential and representative, rather than redundant.

Once a new batch of samples $\mathcal{B}$ is selected and labeled by experts, CIFAD updates its model parameters using the expanded labeled set and then recalculates the influence scores based on the updated network. This process forms a closed-loop training paradigm in which the model is first trained on the available labeled data, then estimates the utility of remaining unlabeled samples, selects the most informative and diverse ones for annotation, and finally retrains with the newly labeled instances. By repeatedly cycling through these stages, CIFAD progressively refines its understanding of abnormal behavior, enhances detection accuracy, and reduces annotation overhead under limited supervision.

Overall, this active annotation strategy bridges the gap between model-driven inference and human-in-the-loop supervision. By systematically prioritizing informative and diverse samples, CIFAD adaptively expands its labeled set and maintains stable performance even in extremely low-label social network environments. The resulting loop not only makes more effective use of scarce labeled data but also enables practical deployment in dynamic monitoring systems where annotation budgets are limited.

4.6 Overall Objective Function

To jointly optimize all components of CIFAD, we integrate the objectives from the encoder, causal decomposition, and classifier into a unified loss function. This total objective combines the supervised classification term with multiple regularization components that promote invariance, disentanglement, and sparsity, ensuring accurate and stable anomaly detection in dynamic social networks.

Formally, the overall loss is defined as:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{IRM}} \mathcal{L}_{\text{IRM}} + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}} + \lambda_{\perp} \mathcal{L}_{\perp} + \lambda_{\text{spa}} \mathcal{L}_{\text{spa}}. \quad (17)$$

Here, $\mathcal{L}_{\text{cls}}$ denotes the supervised classification loss that drives the anomaly detector to distinguish between normal and abnormal nodes. $\mathcal{L}_{\text{IRM}}$ enforces invariant risk minimization, guiding the model toward causal representations that generalize across pseudo-environments. $\mathcal{L}_{\text{adv}}$ originates from the environment discriminator and encourages the causal subspace to remain independent of environmental noise. The orthogonality constraint $\mathcal{L}_{\perp}$ ensures that the causal and spurious subspaces capture distinct, non-overlapping information, preventing leakage between them. Finally, the sparsity term $\mathcal{L}_{\text{spa}}$ regularizes the attention gates in the encoder, encouraging focus on the most informative relation-lag dependencies while suppressing redundancy. The coefficients λ_{IRM} , λ_{adv} , $\lambda_{\perp}$, and λ_{spa} control the relative contribution of each term and are empirically tuned using validation data.

This joint optimization balances discriminative accuracy and causal consistency. While $\mathcal{L}_{\text{cls}}$ provides direct supervision, the regularization terms collectively enhance structural disentanglement, environmental robustness, and interpretability. Through this unified objective, CIFAD achieves reliable few-shot anomaly detection by effectively separating stable causal signals from transient contextual noise.

5 Experiments

5.1 Experimental Settings

1) *Datasets*. We conduct experiments on five representative real-world social network datasets to evaluate the effectiveness and

Table 1: Statistics of Datasets

Dataset	OTC	Weibo	Reddit	YelpChi	Twitter
# Nodes	5,858	8,405	232,965	45,954	17,069,982
# Edges	35,592	407,963	114,615,892	144,584	71,835,017
# Features	4	500	602	32	3
# Time Steps	60	90	12	12	7
# Node Types	3	4	1	3	1
# Edge Types	4	6	1	3	1
Lag Window	5	5	5	5	5
# Anomalies	178	868	12,818	6,480	341,400
Anomaly (%)	3.0	10.3	5.5	14.1	2.0
Avg. Degree	6.1	48.5	492	6.3	8.4
Train Steps	42	63	8	8	5
Test Steps	18	27	4	4	2

generality of CIFAD. As summarized in Table 1, these datasets exhibit diverse scales and properties. **Weibo** and **Twitter** capture dynamic microblogging interactions with anomaly rates of 10.3% and 2.0%, respectively. **Reddit** represents dense community networks with an average degree of 492, while **Bitcoin-OTC** provides a trust network containing heterogeneous relations. **YelpChi** offers review-based social data and shows the highest anomaly proportion at 14.1%. Following prior work [6, 19, 32, 36], we preprocess all datasets into temporal snapshots and use 0.5%–2% labeled anomalies for few-shot evaluation.

2) *Evaluation Metrics*. We employ two widely used metrics in anomaly detection: AUC and Recall@ k . AUC (Area Under the ROC Curve) measures the probability that the model assigns a higher anomaly score to an anomalous instance than to a normal instance. It provides a robust evaluation under imbalanced class distributions and is insensitive to threshold selection, which makes it a standard ranking-based metric adopted by most existing works. In addition, we report Recall@ k , which quantifies the proportion of true anomalies successfully identified within the top- k instances ranked by their anomaly scores. Here, k equals the total number of ground-truth anomalies in the dataset. Recall@ k directly reflects the model’s ability to capture the most critical anomalies, which is particularly important in high-risk scenarios such as fraud detection or security monitoring, where missing an anomaly can lead to severe consequences. Together, AUC and Recall@ k provide a comprehensive evaluation of both global ranking performance and top-ranked detection effectiveness.

3) *Baselines*. To comprehensively assess the performance of CIFAD, we compare it against six representative baseline methods covering static, dynamic, and few-shot graph anomaly detection paradigms. All baselines are re-implemented or adapted to our dynamic few-shot setting, using consistent node features, sliding-window temporal segmentation, and unified sampling protocols for fair comparison.

- **GRAM** [32] is an interpretable static graph anomaly detection model that leverages gradient attention maps to locate anomalous regions within graphs. By highlighting the contribution of individual nodes and edges to anomaly scores, GRAM enhances transparency and serves as a representative static baseline.
- **UMGAD** [10] is an unsupervised multiplex graph anomaly detection method that employs a graph-masked autoencoder

Table 2: Comparison of CIFAD and baselines under different few-shot settings on five dynamic social network datasets. Both AUC and Recall@k are reported. Best results are in bold.

Metric	Method	Weibo			Twitter			Reddit			OTC			YelpChi		
		1-shot	5-shot	10-shot	1-shot	5-shot	10-shot	1-shot	5-shot	10-shot	1-shot	5-shot	10-shot	1-shot	5-shot	10-shot
AUC	GRAM [32]	0.585	0.610	0.625	0.570	0.600	0.615	0.620	0.645	0.660	0.560	0.585	0.600	0.590	0.615	0.630
	UMGAD [10]	0.610	0.640	0.655	0.600	0.630	0.645	0.650	0.675	0.690	0.580	0.605	0.620	0.620	0.650	0.665
	RustGraph [6]	0.670	0.705	0.720	0.660	0.690	0.705	0.710	0.735	0.745	0.630	0.660	0.675	0.670	0.700	0.715
	GeneralDyG [31]	0.680	0.715	0.730	0.670	0.700	0.715	0.720	0.740	0.755	0.640	0.670	0.685	0.680	0.710	0.725
	Coft-AD [12]	0.690	0.730	0.740	0.680	0.710	0.725	0.730	0.755	0.765	0.650	0.680	0.690	0.690	0.720	0.730
	KAG-Prompt [18]	0.700	0.745	0.752	0.690	0.720	0.730	0.740	0.760	0.770	0.660	0.685	0.695	0.700	0.725	0.735
	CIFAD (ours)	0.715	0.821	0.836	0.705	0.800	0.820	0.750	0.825	0.840	0.670	0.760	0.778	0.710	0.790	0.808
Recall@k	GRAM [32]	0.780	0.800	0.810	0.760	0.780	0.790	0.800	0.820	0.830	0.740	0.760	0.770	0.770	0.790	0.800
	UMGAD [10]	0.810	0.830	0.845	0.790	0.810	0.825	0.830	0.850	0.865	0.780	0.800	0.815	0.800	0.830	0.845
	RustGraph [6]	0.840	0.860	0.870	0.820	0.840	0.855	0.870	0.885	0.895	0.810	0.830	0.845	0.830	0.860	0.870
	GeneralDyG [31]	0.850	0.865	0.875	0.830	0.850	0.860	0.880	0.890	0.900	0.820	0.840	0.850	0.840	0.870	0.880
	Coft-AD [12]	0.855	0.868	0.878	0.835	0.855	0.865	0.885	0.895	0.905	0.825	0.845	0.855	0.845	0.875	0.885
	KAG-Prompt [18]	0.858	0.875	0.882	0.838	0.858	0.868	0.888	0.898	0.905	0.828	0.848	0.858	0.848	0.878	0.885
	CIFAD (ours)	0.860	0.885	0.895	0.840	0.870	0.880	0.890	0.905	0.915	0.830	0.855	0.865	0.850	0.885	0.895

with contrastive learning to capture multi-relational correlations across heterogeneous graphs. It serves as a strong non-temporal baseline for heterogeneous anomaly detection.

- **RustGraph** [6] introduces a robust anomaly detection framework for dynamic graphs by jointly modeling structural and temporal dependencies. It combines a variational graph autoencoder with fine-grained contrastive learning to alleviate the impact of noisy labels, making it one of the most competitive temporal baselines.
- **GeneralDyG** [31] addresses generalizability issues in dynamic graph anomaly detection by sampling temporal ego-graphs and sequentially extracting both structural and temporal features. It effectively adapts to evolving social network dynamics and serves as a strong general-purpose dynamic baseline.
- **Coft-AD** [12] is a contrastive fine-tuning method for few-shot anomaly detection. It pre-trains a model on large-scale normal data and fine-tunes it on few-shot target domains using cross-instance contrastive learning, significantly improving adaptation under data-scarce conditions.
- **KAG-Prompt** [18] introduces a kernel-aware graph prompt learning method for few-shot anomaly detection. It constructs hierarchical graphs across feature layers to model contextual relationships and employs multi-level fusion for more accurate anomaly localization in few-shot scenarios.

5.2 Main Comparison Results

We evaluate CIFAD and all baseline methods described in Section 5.1 on the five dynamic social network datasets. All models are trained and tested under the same dynamic few-shot protocol using 1-, 5-, and 10-shot settings, which correspond to roughly 0.5%–2% of labeled anomalies. We report both AUC and Recall@k metrics, where k denotes the number of ground-truth anomalies.

As shown in Table 2, CIFAD consistently outperforms all baseline methods across both metrics on all datasets. Compared with the strongest baseline KAG-Prompt, CIFAD achieves an average improvement of about 7% in AUC and 0.7% in Recall@k, demonstrating superior generalization and robust performance under limited supervision under dynamic few-shot settings.

Table 3: Ablation study of CIFAD on the Weibo dataset. Each variant removes one key module from the full model. Best results are in bold.

Model Variant	AUC	Recall@k
Full CIFAD	0.820	0.885
w/o Sparse Lagged Attention	0.765	0.822
w/o Causal Subspace Decomposition	0.738	0.795
w/o Active Annotation	0.770	0.845

Static baselines such as GRAM and UMGAD perform the weakest since they ignore temporal evolution and rely only on static structural correlations, making them less effective in time-varying social interactions. Dynamic models like RustGraph and GeneralDyG better capture temporal dependencies but remain correlation-driven and thus susceptible to transient environmental noise. Few-shot baselines such as Coft-AD and KAG-Prompt improve performance under label scarcity, yet their learned representations lack causal invariance and robustness to dynamic context shifts.

By contrast, CIFAD separates stable causal signals from environmental noise via causal-invariant subspace decomposition and employs an active annotation mechanism to make more effective use of scarce labeled data. This combination leads to consistent gains in both AUC and Recall@k across all datasets, confirming the advantages of causal modeling and active annotation strategy designed for limited supervision.

Overall, CIFAD shows strong accuracy and robustness across dynamic heterogeneous social networks.

5.3 Ablation Study

To examine the role of each major component in CIFAD, we perform ablation studies on the Weibo dataset under the same 5-shot setting. Each time, one module is removed from the full model to observe the change in AUC and Recall@k. The results are reported in Table 3. **Effect of Sparse Lagged Attention.** When the sparse lagged attention (HGTT-SLA) is removed, CIFAD degenerates into a temporal

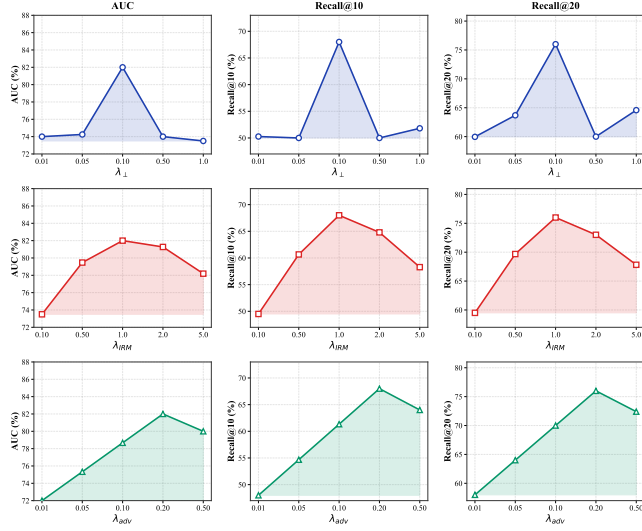


Figure 2: Parameter sensitivity analysis of CIFAD on the Weibo dataset. The three rows correspond to variations of the orthogonality regularization weight $\lambda_{\perp}$, the invariant risk minimization weight λ_{IRM} , and the adversarial deconfounding weight λ_{adv} , respectively. Each column reports AUC, Recall@10, and Recall@20.

GCN that treats all time-lagged edges equally. As shown in Table 3, the performance notably declines in both AUC and Recall@k, indicating that selectively emphasizing informative temporal dependencies is crucial for capturing abnormal user behaviors.

Effect of Causal Subspace Decomposition. Disabling the causal-invariant decomposition results in a representation that entangles stable causal signals with transient noise, leading to overfitting to environmental variations. This variant exhibits the largest performance drop, especially in Recall@k, verifying that disentangling stable causal mechanisms enhances robustness and generalization under dynamic social conditions.

Effect of Active Annotation. Replacing the influence-function-based active annotation module with random sampling leads to a noticeable performance drop, especially in the few-shot setting. Without actively choosing informative samples, the model needs more labeled data to reach comparable accuracy, confirming that the active annotation strategy makes more effective use of scarce labeled data.

Overall, the ablation results demonstrate that each module of CIFAD contributes significantly to the final performance, and the combination of causal-invariant representation learning with the active annotation strategy is the main factor behind its superiority.

5.4 Parameter Sensitivity Analysis

To evaluate the robustness of CIFAD, we conduct a parameter sensitivity analysis on three key hyperparameters: the orthogonality regularization weight $\lambda_{\perp}$, the invariant risk minimization weight λ_{IRM} , and the adversarial deconfounding weight λ_{adv} . Experiments are performed on the Weibo dataset under the 5-shot setting, and results are summarized in Figure 2.

The first row of Figure 2 studies $\lambda_{\perp}$, which controls the strength of the orthogonality constraint between the causal and spurious subspaces. Increasing $\lambda_{\perp}$ from 0.01 to 0.1 consistently improves both AUC and Recall@k, indicating that moderate orthogonality regularization facilitates disentanglement and reduces spurious interference. However, when $\lambda_{\perp}$ exceeds 0.5, performance degrades, suggesting that over-regularization may overly restrict representation capacity and hinder the model from capturing informative temporal dynamics.

The second row reports the effect of λ_{IRM} , which enforces environment invariant prediction across pseudo-environments. Performance improves as λ_{IRM} increases from 0.1 to 1.0, showing that invariant regularization benefits robustness under distribution shifts. Beyond this range, further increasing λ_{IRM} leads to a gradual decline, likely because excessively strong invariance constraints suppress environment-specific patterns that remain predictive for anomaly detection.

The third row presents the sensitivity to the adversarial weight λ_{adv} , which encourages the representation to be insensitive to environment-related confounders via adversarial training. We observe a clear performance gain as λ_{adv} increases from 0.01 to 0.05, while larger values bring marginal improvements and may even plateau, implying that moderate adversarial deconfounding is sufficient to reduce spurious correlations without removing useful causal signals.

Overall, CIFAD achieves stable performance when $\lambda_{\perp} = 0.1$, $\lambda_{IRM} = 1.0$, and $\lambda_{adv} = 0.05$, demonstrating robustness to reasonable hyperparameter variations. These results further confirm that each regularization component contributes meaningfully to reliable anomaly detection in dynamic heterogeneous graphs.

6 Conclusion and Future Work

This paper presents CIFAD, a causal-invariant few-shot anomaly detection method for dynamic social networks. By integrating causal subspace decomposition with an active annotation loop, CIFAD effectively enhances generalization under distribution shifts and makes more effective use of limited labeled data compared with existing methods. The method further provides interpretable and stable representations for real-world social monitoring scenarios. Extensive experiments across multiple datasets validate the superiority of CIFAD and demonstrate how each component contributes to its performance and robustness.

In future work, we aim to develop automatic environment discovery and adaptive invariant modeling, enabling the method to autonomously identify shifting causal factors without explicit environment labels.

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